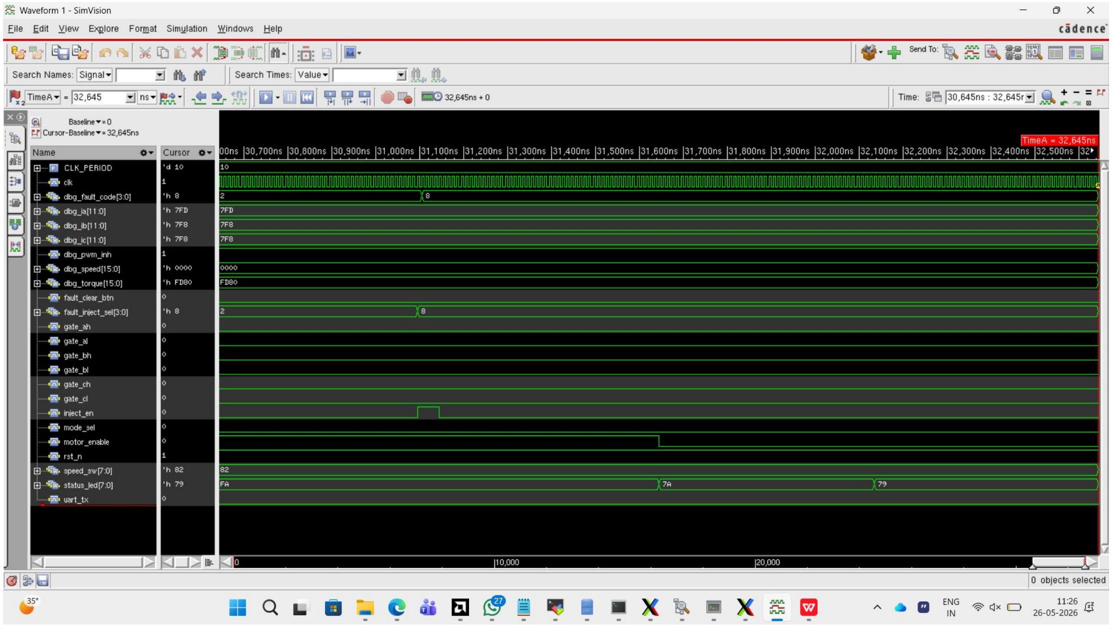
NC Launch Simulation



Clock & Reset

- clk — toggling correctly at 10 ns period (100 MHz) — simulation clock is healthy
- rst_n = 1 — reset has been de-asserted, system is running

Motor State

- motor_enable — goes HIGH then LOW partway through, matching **Test 9 (motor disable)** in your testbench
- mode_sel = 0 — V/Hz sinusoidal mode is active
- speed_sw[7:0] = 0x82 (130 decimal) — speed command sent to the DFC

Fault Injection & Protection

- fault_inject_sel[3:0] transitions **2 → 8** — this is your testbench running:
  - 0x2 = Overvoltage injection (Test 7)
  - 0x8 = Short-circuit injection (Test 8)
- dbg_fault_code[3:0] transitions **2 → 8** correspondingly — the protection unit **correctly latched and encoded** each fault
- dbg_pwm_inh = 1 — PWM inhibit is **active**, meaning protection_unit fired and blocked all gate outputs
- inject_en — short pulse visible around 31,000 ns — this is the injection trigger

Gate Outputs (All Zero)

- gate_ah, gate_al, gate_bh, gate_bl, gate_ch, gate_cl — all **0** throughout this window — correct behaviour because pwm_inhibit = 1 is masking all outputs during the fault state

Current & Speed Signals

- dbg_ia[11:0] = 0x7FD, dbg_ib = 0x7F8, dbg_ic = 0x7F8 — these are **mid-rail ADC values (~2048 counts)** which is exactly correct — the virtual motor model outputs current as unsigned 12-bit centred at 2048, meaning near-zero real current
- dbg_speed[15:0] = 0x0000 — rotor speed is zero, consistent with motor being disabled and PWM inhibited
- dbg_torque[15:0] = 0xFD80 — a non-zero torque estimate residual from the model state

Status LED

- status_led[7:0] transitions **0xFA → 0x7A → 0x79** — decoding the 8-bit mapping from your code:

Bit	Signal	FA	7A	79
[7]	motor_enable	1	0	0
[6]	fault_any	1	1	1
[5]	fi_active	1	1	1
[4]	hall_valid	1	1	1
[3]	zc_detected	1	1	0
[2:0]	hall_sector	010	010	001

Motor disabled at the FA→7A transition, rotor sector changing at the end.

UART

- uart_tx = 0 — transmit line is low (idle mark = 1, so this may indicate a start bit or the line is held — check if telemetry streaming is active)
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Overall Assessment ✅

simulation is **passing correctly**:

- Fault injection triggers protection as expected
- PWM inhibit correctly kills all gate signals
- Fault codes match injected fault types
- Status LED reflects system state changes
- Motor disable sequence is visible